Dongsuk Jang

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Education

Seoul National University, Seoul, Korea | MedInfoLab — Mar 2023 – Feb 2026
M.S. in Interdisciplinary Program in Bioengineering (Advisor: Jinwook Choi)

Yale University, New Haven, CT, USA | YaleNLP — Oct 2024 – Sep 2025
Visiting Student, Department of Computer Science (Advisor: Arman Cohan)

Sungkyunkwan University, Suwon, Korea — Mar 2015 – Feb 2023
B.S. in Integrative Biotechnology (Micro Degree in Core-Bio A-School Track)

Publications

[1] Jaeyoung Kim, Eunseok Kim, **Dongsuk Jang**, Seunghyun Kim. Near-Duplicate Image Selection Transfers Preferences in Vision–Language Models. Under review at ICLR 2027.

[2] Jaeyoung Kim, Eunseok Kim, **Dongsuk Jang**. Is the Geometry Doing the Work? An Operating-Point Audit of Hierarchy in Hyperbolic Vision-Language Models. TMLR, 2026. [PDF]

[3] **Dongsuk Jang**, Kyle Tegtmeyer, Muhammad Qureshi, Anurag Gupta, Ziyao Shangguan, Sophie Chheang, Arman Cohan. Retrieval-Augmented Structured Reporting for Radiology Workflow Integration. SIIM-CAIMI 2025.

[4] **Dongsuk Jang**, Ziyao Shangguan, Kyle Tegtmeyer, Anurag Gupta, Jan Czerminski, Sophie Chheang, Arman Cohan. MedTutor: A Retrieval-Augmented LLM System for Case-Based Medical Education. EMNLP 2025 System Demo. [PDF]

[5] **Dongsuk Jang**, Alan Li, Arman Cohan. Multi-Perspective Integration via Mixture-of-Agents for Enhanced Healthcare QA Summarization. CL4Health Workshop @ NAACL 2025. [PDF]

[6] Hyeryun Park, Jeongwon Kwak, **Dongsuk Jang**, Sumin Park, Jinwook Choi. LoraMap: Harnessing the Power of LoRA Connections. Under review at IEEE Journal of Biomedical and Health Informatics. [PDF]

[7] **Dongsuk Jang**†, Hyeryun Park†, Jiye Son, Hyeonuk Hwang, Su-jin Kim, Jinwook Choi. Automated Information Extraction from Thyroid Operation Narrative: A Comparative Study of GPT-4 and Fine-Tuned KoELECTRA. AMIA Informatics Summit 2024. [PDF]

Projects

Medical RAG Failure Analysis & Evidence Selection Benchmark — Apr 2026 – Present
Constructing a medical RAG benchmark in collaboration with clinicians to evaluate LLM failure modes. Identified key limitations of existing reranking approaches in medical contexts and formulated a novel framework establishing the necessity of an explicit evidence set selection stage.

Evidence-Aware Selector for Reasoning Intensive Medical RAG — Apr 2026 – Present
Developing an evidence-aware selector model tailored for reasoning intensive medical tasks. Designed a specialized model architecture based on evidence set selection, significantly improving retrieval relevance and mitigating LLM hallucinations in clinical RAG pipelines.

Clinical NL2SQL Agent Evaluation for Large-Scale Hospital Databases — Apr 2026 – Present
Analyzing the feasibility of NL2SQL agents for retrieving clinically relevant information or constructing patient cohort from tertiary hospital CDW and EMR systems, with planned expert evaluation in real-world Oracle database environments.

Table-to-Text Generation for Clinical Decision Support [Link] — Dec 2023 – Jul 2024
Led a full-cycle project to mitigate clinician cognitive load by generating natural language summaries from complex prescription tables. Directed the creation of a ground-truth dataset using Seoul National University Hospital data and systematically evaluated traditional Seq2Seq architectures against modern LLMs fine-tuned with PEFT.

Automatic Generation of Thyroid Operation Records on Web [Video] — Mar 2023 – Feb 2024
Led the model development lifecycle to automate the creation of structured thyroid operation records from voice-recorded surgical notes. Trained and rigorously evaluated a fine-tuned KoELECTRA model to ensure high-accuracy clinical information extraction.

Evaluating Transformers for Clinical Natural Language Inference [GitHub] — Dec 2022 – Feb 2023
Conducted a comparative performance analysis of various models, including m-LSTM and multiple BERT-based architectures, on the MedNLI task with the MIMIC-III dataset. Identified and implemented optimal parameter settings to achieve the best performance.

Experience

Student Intern, Samsung Advanced Institute for Health Sciences and Technology Mar 2021 – Dec 2022
Acquired foundational skills in Python and Natural Language Processing, applying them to conduct preliminary research on analyzing biomedical text data under expert mentorship.

Military Service, Korean Augmentation to the U.S. Army (KATUSA) Jan 2018 – Sep 2019
Managed human resources and administrative operations for a multinational military unit, developing strong organizational and cross-cultural communication skills.

Honors and Awards

BK21 Research Scholarship, Seoul National University 2025
A competitive, government-funded scholarship for outstanding graduate students.

2nd Place in PerAnsSumm Shared Task at CL4Health @ NAACL [Link] 2025

Biomedical Global Talent Nurturing Program [Link] 2024
Awarded a $30,000 grant by the Korea Health Industry Development Institute for visiting research opportunity.

2nd Place, Seocho YES 5th Startup Pitching Competition [Link] 2024
Selected for the competitive startup incubation program and secured grant funding for project development.

Selected Team, 6th SNU Haedong Junior Startup Program 2024

Next Generation BioHealth Leader Award, Sungkyunkwan University 2023

Best Research Intern Award, Samsung Advanced Institute for Health Sciences and Technology 2021

Commandant's List, KATUSA Training Academy 2018

Professional Services

Reviewer for the Oct ARR 2025 2025

Lead Graduate Student, MEDINFO Lab, Seoul National University Sep 2025 – Feb 2026

Local organizer for NENLP 2025 [Link] 2025

Reviewer for the AMIA 2025 Annual Symposium, Informatics Summit 2024

Skills Summary

Languages
Python, SQL, R, Matlab

NLP Libraries
PyTorch, Transformers, vLLM, Accelerate, bitsandbytes, LangChain

References

Arman Cohan, Ph.D.
Assistant Professor, Department of Computer Science, Yale University
arman.cohan@yale.edu

Jinwook Choi, M.D., Ph.D.
Professor, Interdisciplinary Program in Bioengineering, Seoul National University
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Seong-Hwan Kim, M.D., Ph.D.
Assistant Professor, Department of Orthopaedic Surgery, Chung-Ang University Hospital
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